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SUPERSEDING
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31 MAY 1967

MILITARY STANDARD

FASTENER TEST METHODS

METHOD 4,

LAP JOINT SHEAR

FSC 53GP

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 4, Lap Joint Shear

MIL-STD-1312-4

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

This standard sets forth a standard test method to simulate shearing action on a fastener to determine its shear withstanding capabilities.

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1. SCOPE

1.1 Applicability. This test method outlines a standard procedure for determining the room temperature strength properties of mechanically fastened sheet metal lap joints statically loaded to produce shear on the fastener. The significant strength properties include the ultimate strength and the yield strength of the joint.

1.1.1 This test method applies to sheet metal lap joints fastened with rivets, bolts, screws, or comparable fastening devices.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-45662	Calibration System Requirements

HANDBOOKS

MILITARY

MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 4	Load Verification of Testing Machines
ASTM E 8	Tension Testing of Metallic Materials
ASTM E 83	Verification and Classification of Extensometers

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AEROSPACE INDUSTRIES ASSOCIATION, INC. (AIA)

NAS 618 Fastener, Recommended Shank, Hole and Head to Shank Fillet Radius, Limits for

(Application for copies should be addressed to the Aerospace Industries Association, Inc., 1725 DeSales Street, NW, Washington, DC 20036.)

3. DEFINITIONS

3.1 Joint ultimate shear strength. The ultimate strength of a joint is the highest load developed by the joint coincident with or prior to the failure of the fastener, parent sheet material, or a combination of both.

3.2 Joint yield shear strength. Yield strength of a joint is the load corresponding to a specified value of permanent set across the joint. Unless otherwise specified, the yield shall be determined from the load deflection curve by the Secondary Modulus Method as illustrated in figure 1 using an offset of 0.04D or as defined in MIL-HDBK-5, Section 9.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Testing machine. The testing machine shall be capable of applying a tensile load at a controllable rate. The calibration system for the machine shall conform to MIL-STD-45662. Its accuracy shall be verified every 12 months by a method complying with ASTM E 4. The yield loads, ultimate loads and structural failure loads of the fasteners tested shall be within the loading range of the testing machine as defined in ASTM E 4.

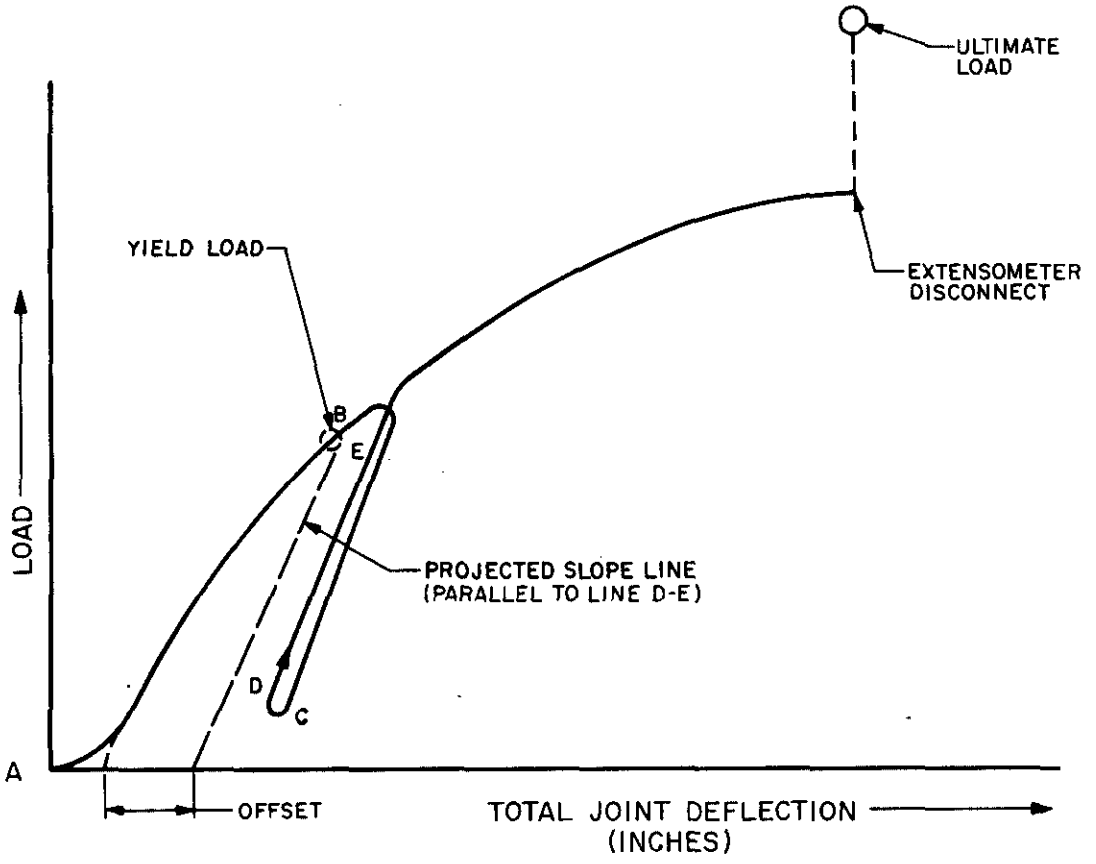
4.1.2 Deflection measuring device. The measuring device used shall be an averaging, differential-transformer extensometer or equivalent, preferably of the separable type. It shall be used in conjunction with an autographic recorder and it shall have an accuracy of 0.5 percent of the indicated joint deflection at yield strength and be calibrated per ASTM E 83. It shall be capable of direct attachment or mounting to the joint specimen. Load and deflection ranges shall be used that give the initial part of the load-extension curve a slope between 45 and 60 degrees.

4.1.3 Test fixtures. The test fixtures holding the joint specimen shall have self-aligning jaws (pin or grip type) capable of transmitting a load along a faying surface of the specimen within 0.005-inch of the centerline of the fasteners.

4.2 Test specimens.

4.2.1 Configuration.

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Typical load deflection curve for lap joint specimen. Curve shall be developed by loading specimen starting at point "A," to vicinity of approximate yield as noted by point "B." The load shall then be relieved to approximately 20 percent of anticipated load as marked by the path going from "B" to "C." The joint specimen shall be reloaded to define the secondary modulus loop as shown by the curve "C"-"D"-"E." In developing the secondary modulus, selection of load drop-off should be beyond the anticipated joint yield load. After the curve has been sufficiently developed to ensure subsequent determination of the joint yield load, the extensometer may be removed, and the joint loaded to ultimate failure. In establishing the joint yield load from the curve, the slope of the secondary modulus (line connecting points "D" and "E") is projected from the specified offset as established for the specific fastener type and size under evaluation. The point where the projected slope line intersects the load deflection curve is defined as the actual joint yield load.

FIGURE 1. Secondary modulus method for determining yield load deflection curve.

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4.2.1.1 Preferred specimen configuration. Unless otherwise specified, the test specimen configuration shall be as shown on figure 2.

4.2.1.2 Alternative specimen configuration. The alternative test specimen configuration, as shown on figure 3, may be used where specified.

4.2.1.3 Method for loading. The configuration of the ends of the joint specimen outside the lap area is optional. Certain parent sheet materials may be of relatively low strength and thin enough to permit satisfactory gripping in friction-type holding fixtures capable of meeting the requirements of 4.1.3. However, for higher strength parent sheet materials where grip slipping may be encountered, and for specimens too thick to meet 4.1.3, the use of pin loading holes is recommended. When pin loading holes are used, they shall be located so that the load will pass through the centerline of the fastener holes within 0.005-inch, as indicated in figures 2 and 3. When required, doublers as shown on figures 2 and 3 are permitted.

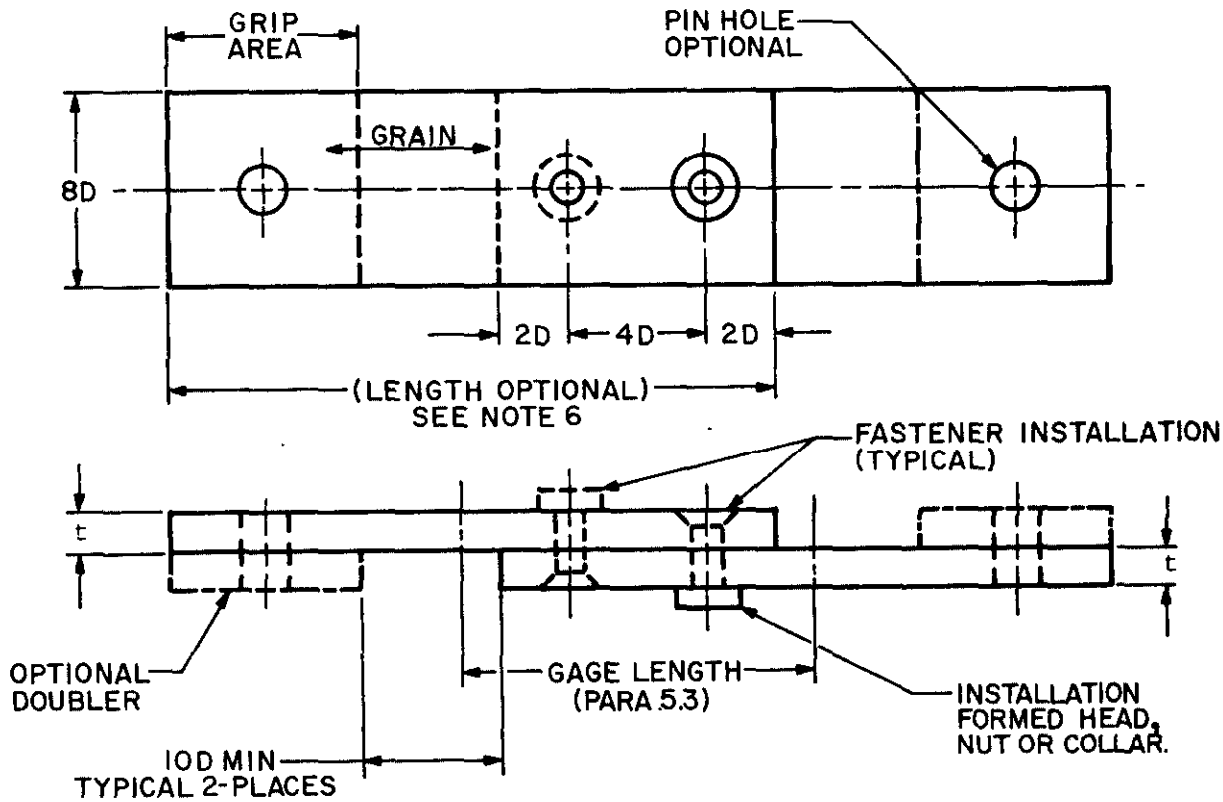
4.2.1.4 Edge distance. Standard tests of lap joint specimens shall be based on specimens maintaining an edge distance of 2D, where "D" equals the nominal diameter of the fastener under evaluation. When the effect of edge distance is to be investigated as a variable, it is permissible to change the 2D dimension.

4.2.2 Preparation.

4.2.2.1 Parent material. Both parent material straps of the test specimen shall be of the same gage and material. The actual thickness of each strap shall be measured adjacent to the lap joint area. Dimensions and grain direction of parent material straps shall be in accordance with figure 2 or figure 3, as applicable. When the effect of grain direction is to be investigated as a variable, it is permissible to use alternate grain directions provided it is so noted in the test report with a drawing accompanying the test report specifically delineating grain direction.

4.2.2.1.1 Joint material mechanical properties. To correlate the data developed under this test method, a minimum of three samples of each parent sheet or plate of the material employed in the actual fastener joint strength evaluation shall be tested for tensile strength. Test procedure and method for determination of tensile properties shall be in accordance with ASTM E 8. The grain direction will be the same as required by the material procurement specification.

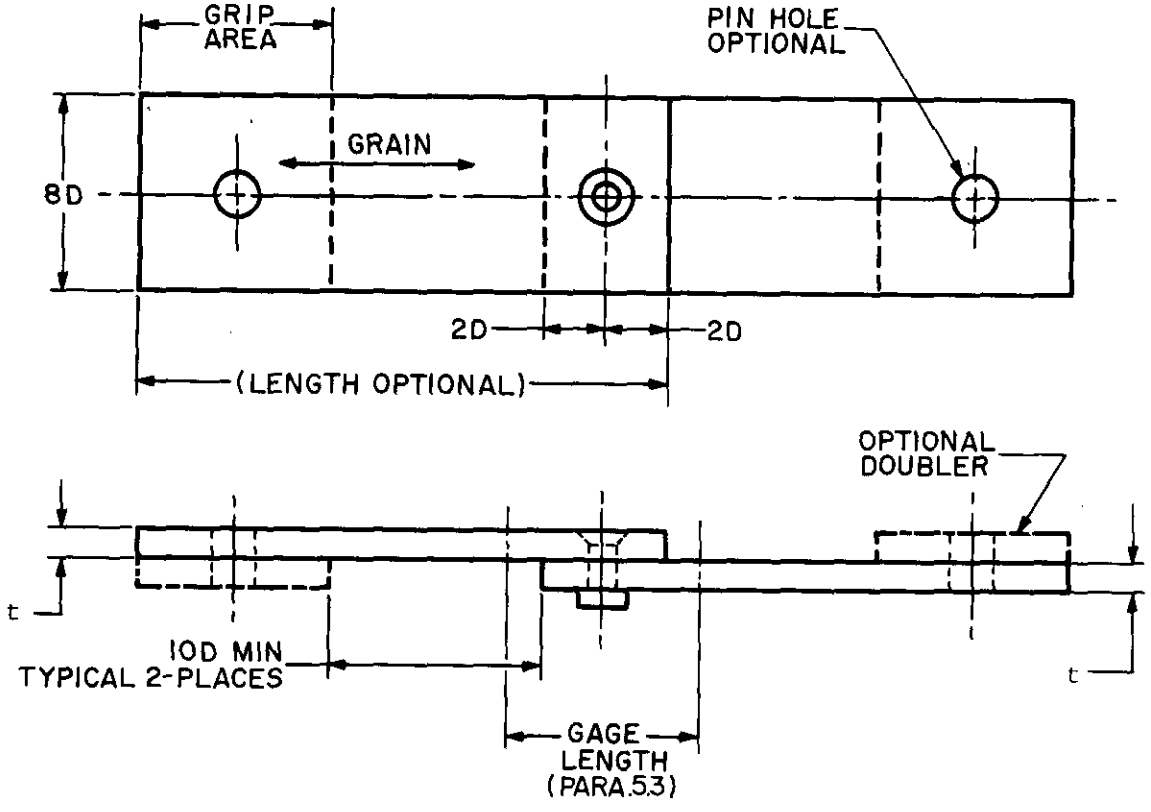
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NOTES:
1. D = Nominal fastener diameter.
2. t = Sheet or plate thickness.
3. Fastener holes, see paragraph 4.2.2.2.
4. Tolerance on edge margin = ± 0.010, spacing and width = ± 0.030.
5. Standard edge distance = 2D.
   For permissible deviations, see paragraph 4.2.1.4.
6. If the optional doubler is used, it shall be attached with a minimum of four fasteners. The fasteners shall have a yield strength greater than half the ultimate strength of the test specimen. If the optional doubler is not used, the specimen length shall not be less than 4D + 5t + 2 times the pin diameter.

FIGURE 2. Preferred test lap joint specimen configuration.

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NOTES:
1. D = Nominal fastener diameter.
2. t = Sheet or plate thickness.
3. Fastener holes, see paragraph 4.2.2.2.
4. Tolerance on edge margin = ± 0.010, width = ± 0.030.
5. Standard edge distance = 2D. For permissible deviations, see paragraph 4.2.1.4.
6. If the optional doubler is used, it shall be attached with a minimum of four fasteners. The fasteners shall have a yield strength greater than half the ultimate strength of the test specimen. If the optional doubler is not used, the specimen length shall not be less than 4D + 5t + 2 times the pin diameter.

FIGURE 3. Optional test lap joint specimen configuration.

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4.2.2.2 Fastener holes. Fastener holes shall be line drilled perpendicular to the sheet surface within 1 degree. When using the test specimen as shown on figure 2, the manufactured heads of the fasteners shall be on opposite sides of the specimen on the 2D edge margin ends of the lap joint. When countersinking is required, the countersink shall be concentric with the hole within 0.002 FIM and the depth shall be such that the installed fastener is flush within +0.002-inch, -0.005-inch. The holes for fasteners shall be relieved on the head side the minimum amount necessary to clear the fastener head-to-shank fillet radius. Hole diameters for specimens containing non-interference shank type fasteners shall be as listed in table I. For standard size solid rivets, hole drill sizes shall be as listed in MIL-HDBK-5, table 8.1.2a. For blind, oversize, taper shank, or other type special fasteners, use hole size limits specified for product application.

TABLE 8.1.2(a). Standard rivet-hole drill sizes and nominal hole diameters.

Rivet size, in.	1/16	3/32	1/8	5/32	3/16	1/4	5/16	3/8
Drill No.	51	41	30	21	11	F	P	W
Nominal hole diameter, in.	0.067	0.096	0.1285	0.159	0.191	0.257	0.323	0.386

4.2.2.3 Specimen cleaning. Specimens shall be cleaned with detergent and rinsed with de-ionized water to obtain a water-break-free surface, followed by rinsing with isopropyl alcohol and air or blow drying. No halogenated solvents or abrasives of any type are permitted. Cleaning must be done within one hour of test, shall be processed with a minimum of handling and shall be handled only with clean, lintless cotton gloves after the cleaning operation.

4.2.3 Assembly.

4.2.3.1 Fastener installation. All solid rivets, swaged collar fasteners, and blind fasteners shall be installed to meet applicable product specification requirements. Nuts shall be torqued to the minimum specified value for the particular bolt-nut combination. Test fasteners shall be installed in the "as supplied" condition, i.e., lubrication shall not be applied to or removed from the manufactured test fastener. Slot-head screws shall be installed with the slot oriented parallel to the transverse axis of the specimen.

4.2.3.2 Grip length. All fasteners shall be the correct grip length for the joint under evaluation. For thicker test joints not critical in bearing, up to one thread may be allowed in bearing for tests involving threaded-type fasteners. Nuts shall not engage incomplete threads next to the bolt grip. Where required, a maximum of two compensating washers shall be used under the nut for adjustment.

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TABLE I. Hole limits for non-interference shank type fasteners.

Nominal fastener size	NAS 618 shank type fastener	
	Minimum limit (in.)	Maximum limit (in.)
0.125	---	---
0.164	.164	.168
0.156	---	---
#10	.190	.194
0.190	---	---
0.250	.250	.254
0.313	.3125	.3165
0.375	.375	.379
0.438	.4375	.4425
0.500	.500	.505
0.563	.5625	.5675
0.625	.625	.630
0.750	.750	.757
0.875	.875	.882
1.000	1.000	1.010
1.125	1.125	1.135
1.250	1.250	1.260
1.375	1.375	1.385
1.500	1.500	1.510

NOTE: Blind, taper shank, and other fasteners, use hole size required for product application.

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4.2.3.3 Nonhole-filling fastener joint preload. Bolts, lockbolts, and other nonhole-filling fasteners shall be installed with sufficient bearing load on the fastener to ensure that no joint slippage resulting from the fastener clearance will be recorded as joint deflection. The recommended procedure for installation of bolts is to assemble the straps by running the nuts up until the faying surfaces are held together lightly and apply the lesser of 100 pounds or 10 percent of the anticipated ultimate joint strength, then tighten the nuts to the required torque. Methods used to provide similar preloads in joints fastened with other types of fasteners shall be described in the test report.

5. DETAIL REQUIREMENTS

5.1 Test procedures.

5.1.1 Installation. The specimen shall be installed in the jaws of the test fixtures specified in 4.1.3. The load shall be transmitted along a line passing through the centerline of the faying surface of the specimen within 0.005-inch of the fastener centerline.

5.2 Load rate. Unless otherwise specified, the load rate shall be 100,000, ±10,000 pounds per minute per square inch of fastener shear area to the approximate yield load, after which it may be increased.

5.3 Gage length. For the joint specimens illustrated in figures 2 and 3, the specimen gage length across the fastener lap section test area shall be as shown in table II. The deflection measuring device, maintaining the specified gage length, shall be centered across the specimen lap area within 0.0625-inch.

6. NOTES

6.1 General. The test method outlined permits the uniform testing of joints and is designed to eliminate or minimize many of the variables that otherwise might occur as a function of test technique or method. It is noted though that joint strength test data is unique in that it requires special handling and analysis to be properly used and interpreted. Some product specifications reference requirements for joint strength tests either for qualification or inspection approval of finished fasteners. The requirements of such specifications should be observed in evaluating the results of joint tests in accordance with this test method. However, substantial number of joint tests are conducted for the purpose of developing and establishing load allowable values for design use in accordance with MIL-HDBK-5 or other similar specific application design criteria. Accordingly, where joint strength tests are to be conducted to satisfy the requirements of MIL-HDBK-5, the requirements of Section 9 of that document should be observed.

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6.2 Test report. The test report shall contain the following data:

a. Sheet material designation and condition.

   Exact test mechanical properties of sheet tensile specimens.

b. Grain direction.

c. Designation of fasteners and accessories used.

   Identify lot numbers.

d. Materials and properties of fasteners and accessories when not specified in item c.

e. Measured fastener diameter and hole size.

f. Specimen size including measured thickness, edge distance, and nominal width and configuration (figure 2 or figure 3, as applicable).

g. Model of test machine used and date of calibration.

h. Model of extensometer used and gage length.

i. Loading rate used.

j. Test ultimate load.

k. Recorded load deflection curves, test yield load, yield offset applied, and method of determining.

l. Type of failure.

m. Report of condition, procedures, influencing test data (discrepancies, preload applied, torque value, hole preparation, and protective finish used).

n. Description of washer (when used).

o. Installation procedure.

p. Cause and duration of any interruptions during test.

q. Results of all inspections.

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TABLE II. Gage length.

Nominal fastener diameter (in.)	Gage length (in.)	
	Figure 2 Lap joint specimen	Figure 3 Lap joint specimen
0.125	2	2
0.156	2	2
0.190	2	2
0.250	4	2
0.313	4	2
0.375	4	2
0.438	4	4
0.500	6	4
0.563	6	4
0.625	6	4
0.688	6	4
0.750	8	4
0.813	8	4
0.875	8	4
1.000	12	6
1.125	12	6
1.250	12	6
1.375	12	6
1.500	--	8

Custodians:
Army - AV
Navy - AS
Air Force - 11

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Navy - AS, SH
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Preparing activity:
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